

5 6
3 3 $R_A = 18$
2 2 $R_B = 12$
4 5 $R_C = 26$
5 4 $R_D = 27$
1 1 $R_E = 7$

同小麥的機率分佈完全相同
不完全相同

$$R_i = 3b(k+1)$$

$$(18^2 + 12^2 + 26^2 + 27^2 + 7^2) - 3 \times 6 \times (5+1)$$

$$> \chi^2_{4, 0.05} = 9.488$$

H_0 產

品種的產有顯著差異

$< 0.05 \Rightarrow \text{Reject } H_0$

結論同(a)

h. 0.60

15-9

(11)

Rank				
voter	Rating	Distance	d	d ²
1	9.5	3	4.5	20.25
2	4	7	-3	9
3	3	12	-9	81
4	12	1	11	121
5	10	8	2	4
6	9.5	11	-3.5	12.25
7	6	4	2	4
8	11	2	9	81
9	1	10	-9	81
10	5	9	-4	16
11	9	5.5	3.5	12.25
12	2	5.5	-3.5	12.25

$$a. r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 454}{12 \times (12^2-1)} \approx -0.589$$

$$b. H_0: \rho_s \geq 0, H_a: \rho_s < 0$$

$$n=12, \alpha=0.05, r_0=0.497$$

$$-0.589 \leq -0.497 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ Rating 和 Distance 存在負相關

①

Rank:

Runner	Years	Finish Time	d_i	d_i^2
1	7	1	6	36
2	9	2	7	49
3	3.5	3	0.5	0.25
4	6	4	2	4
5	8	5	3	9
6	5	6	-1	1
7	2	7	-5	25
8	3.5	8	-4.5	20.25
9	1	9	-8	64

$$a. r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} = 1 - \frac{6 \times 208.5}{9(9^2-1)} = -0.9375$$

$$b. H_0: \rho_s = 0, H_a: \rho_s \neq 0$$

$$n=9, \alpha=0.05, r_0=0.683$$

$$r_s = -0.9375 \leq -0.683 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ Years and Finish Time
存在顯著負相關